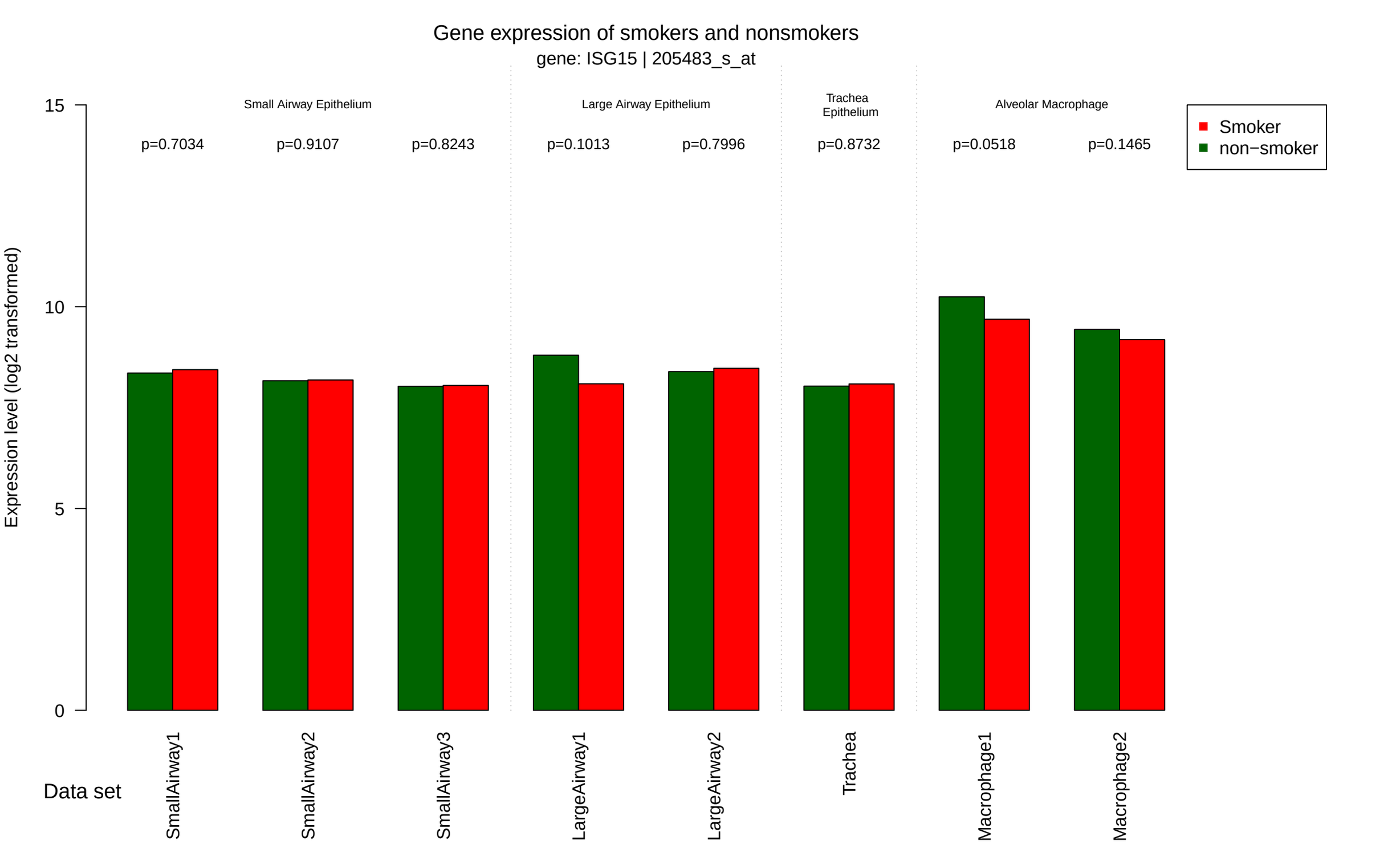
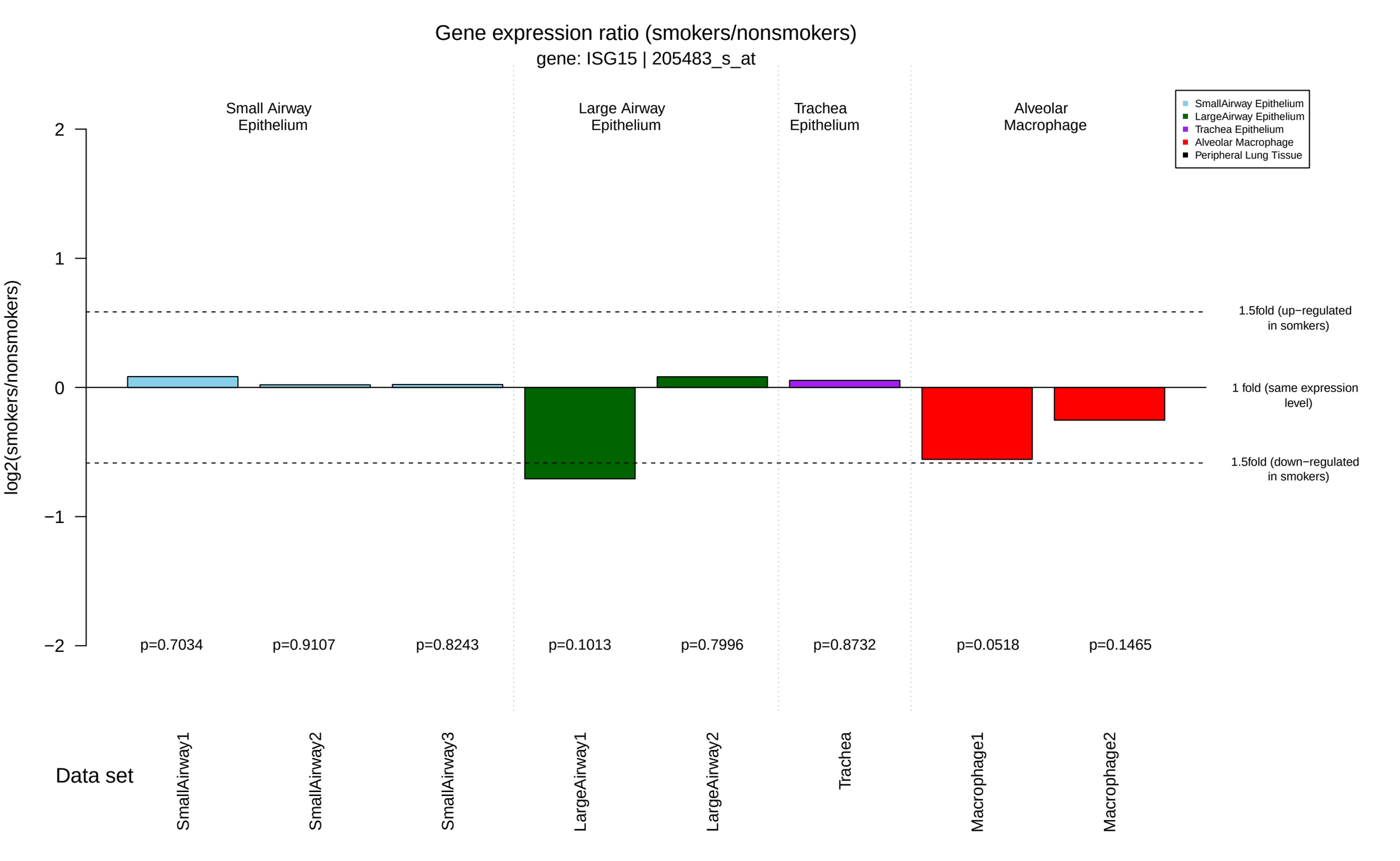
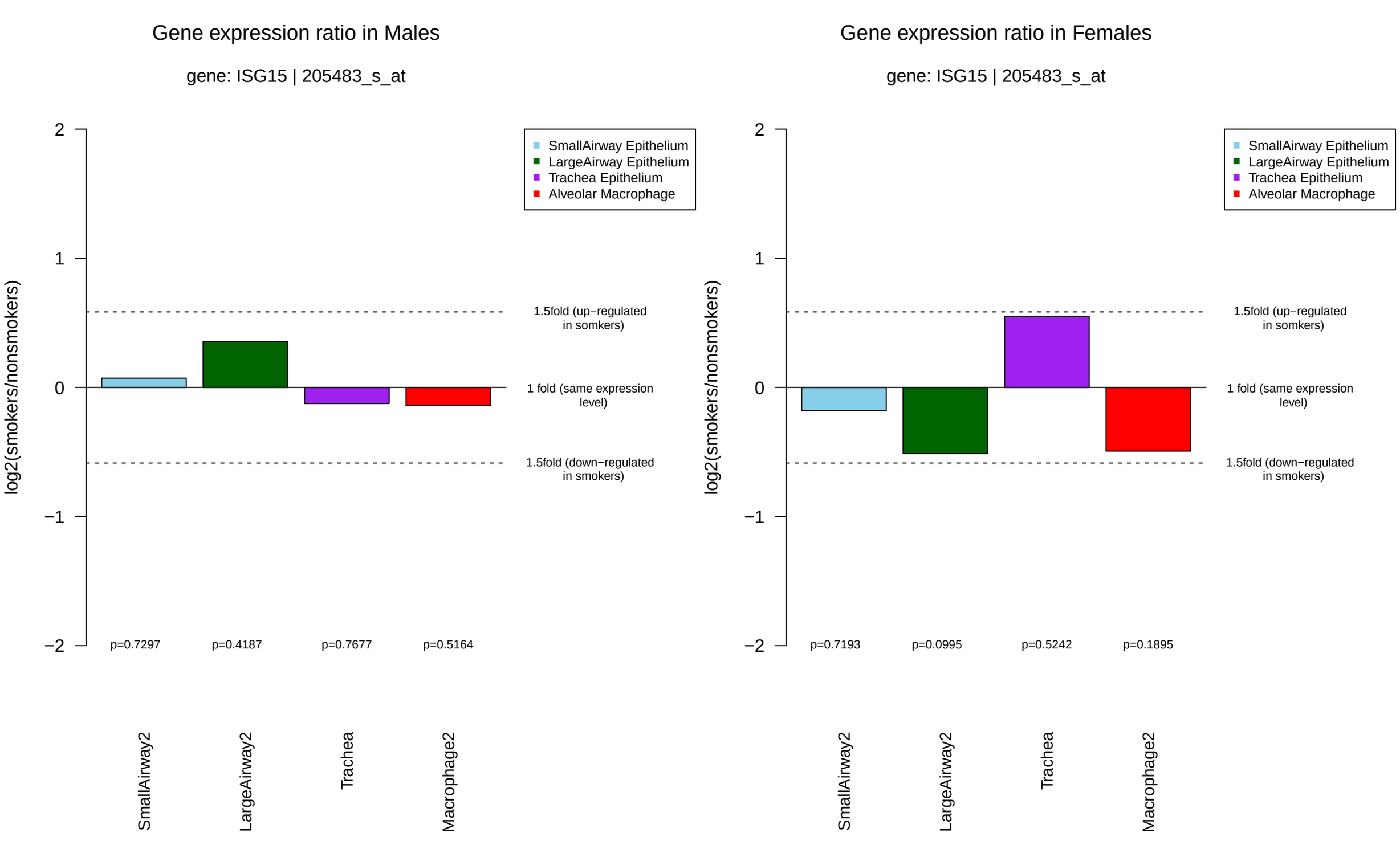
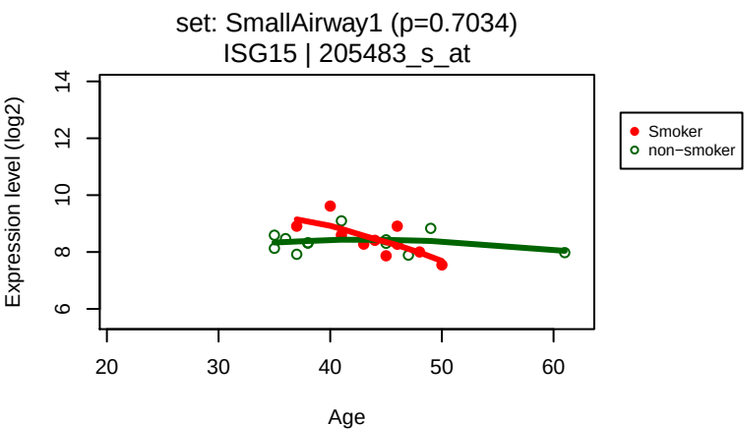
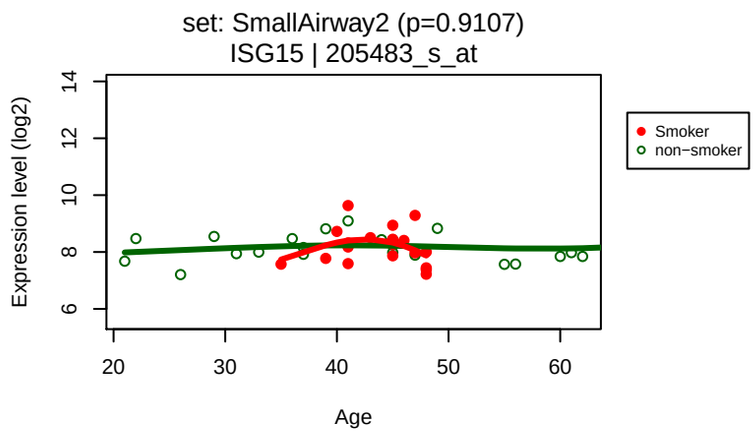
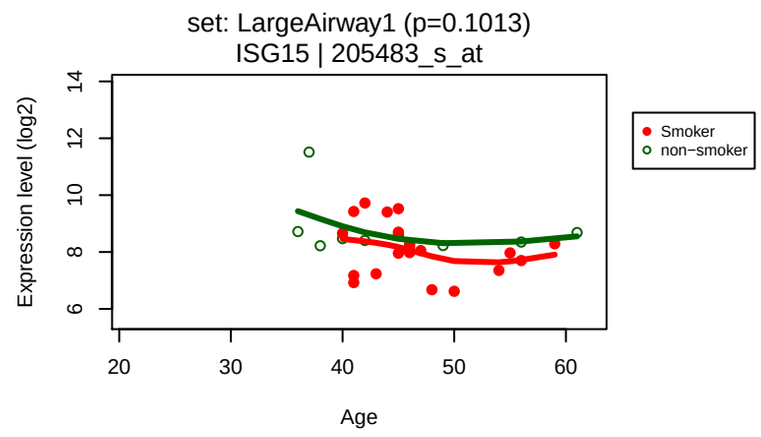

Gene expression of smokers and nonsmokers
gene: ISG15 | 205483_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7034
p=0.9107
p=0.8243
p=0.1013
p=0.7996
p=0.8732
p=0.0518
p=0.1465
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

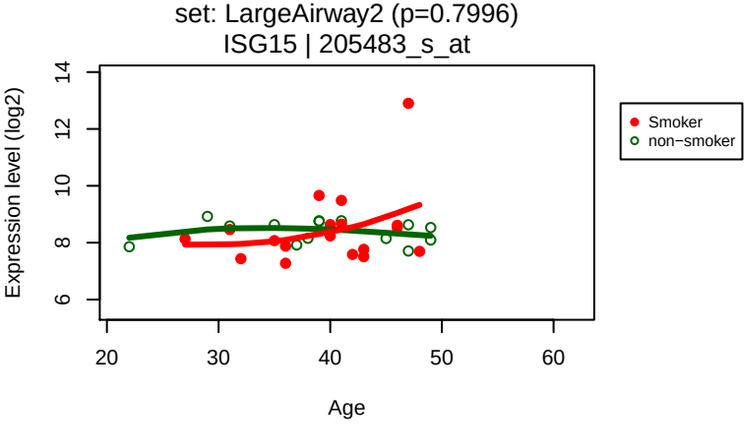

Gene expression ratio (smokers/nonsmokers)
gene: ISG15 | 205483_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7034
p=0.9107
p=0.8243
p=0.1013
p=0.7996
p=0.8732
p=0.0518
p=0.1465
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

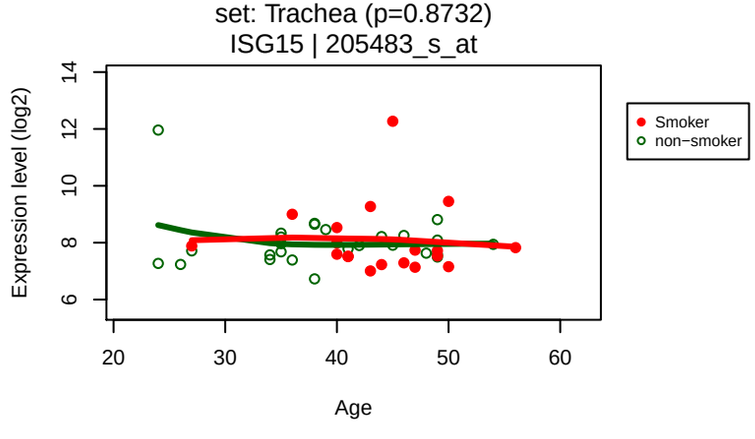

Gene expression ratio in Males
gene: ISG15 | 205483_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7297
p=0.4187
p=0.7677
p=0.5164
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ISG15 | 205483_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7193
p=0.0995
p=0.5242
p=0.1895
SmallAirway2
LargeAirway2
Trachea
Macrophage2

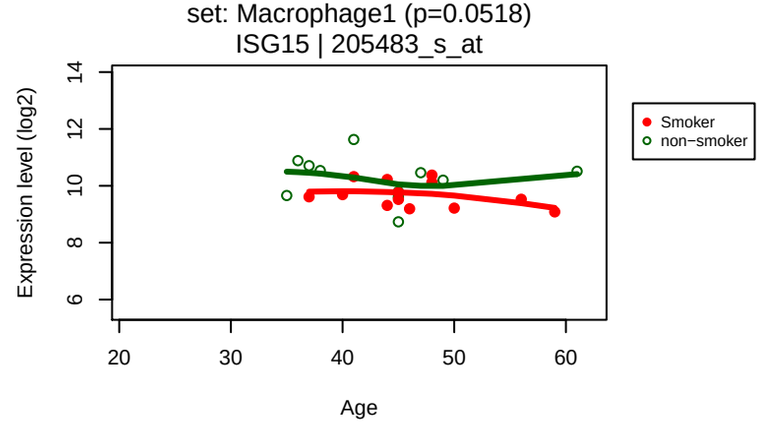

set: SmallAirway1 (p=0.7034)
ISG15 | 205483_s_at
Expression level (log2)
Age
Smoker
non−smoker

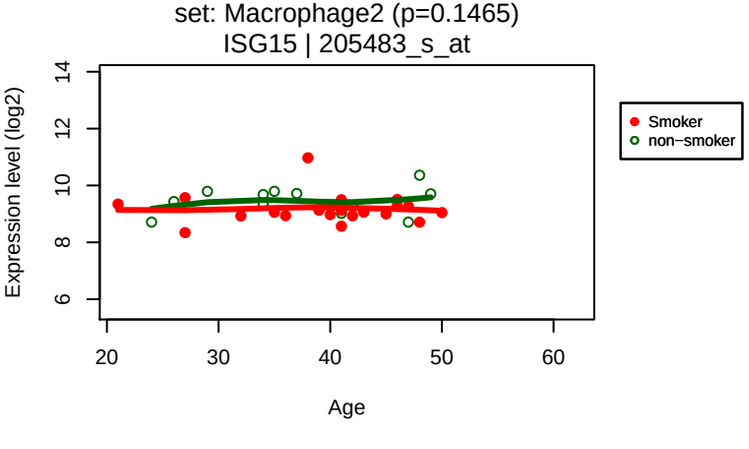

set: SmallAirway2 (p=0.9107)
ISG15 | 205483_s_at
Expression level (log2)
Age
Smoker
non−smoker

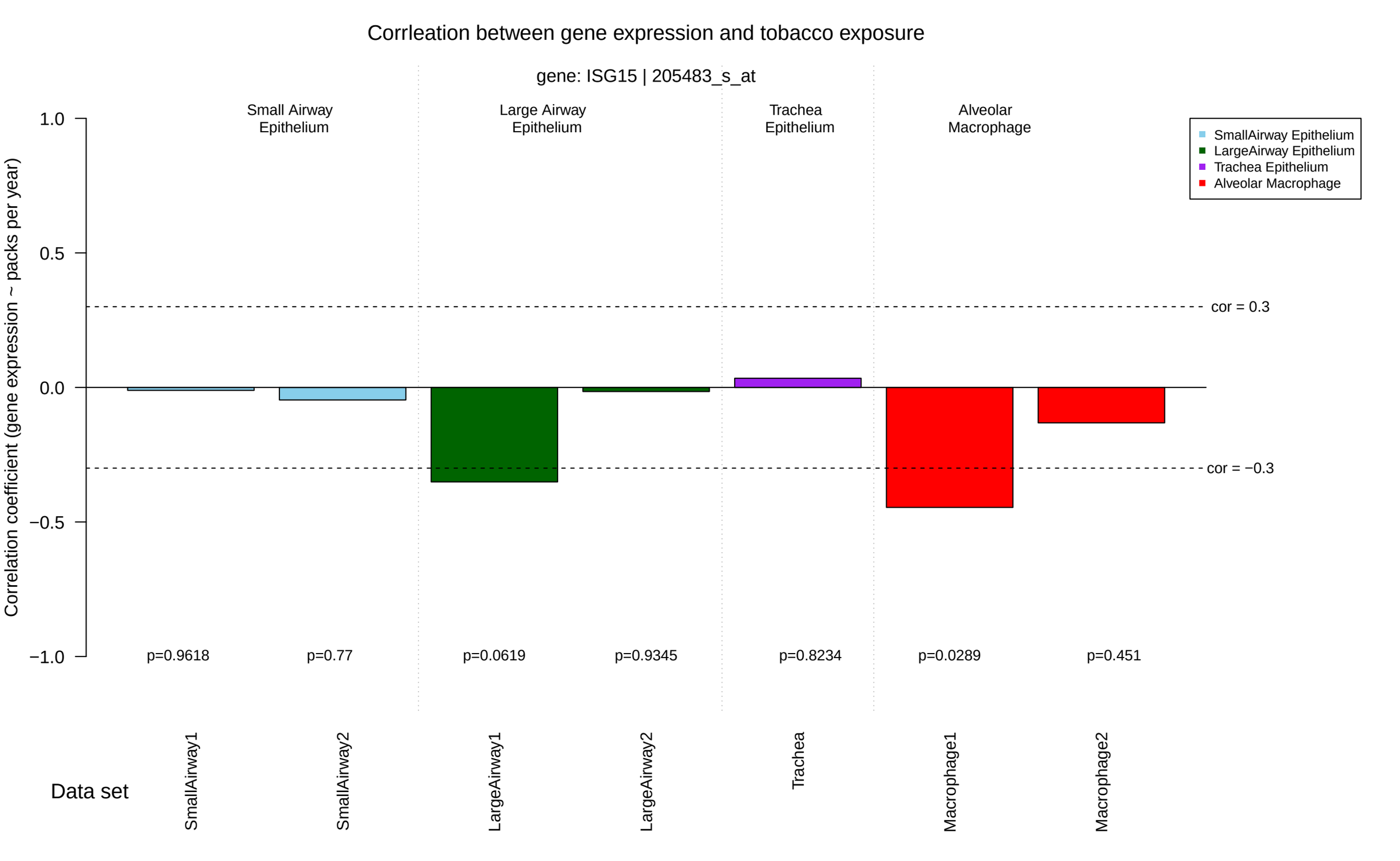

set: LargeAirway1 (p=0.1013)
ISG15 | 205483_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7996)
ISG15 | 205483_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8732)
ISG15 | 205483_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0518)
ISG15 | 205483_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1465)
ISG15 | 205483_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ISG15 | 205483_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9618
p=0.77
p=0.0619
p=0.9345
p=0.8234
p=0.0289
p=0.451
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2